MINI-ASSESSMENT

# Campus Micro-Experiment Designer

A multi-step AI prompt workflow for turning an everyday campus observation into a measurable experiment and an evidence-based action plan.

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**Workflow type**
**Sequential prompt pipeline • 6 stages**

**Core pattern**
**Observe → Decompose → Diagnose → Prioritize → Experiment → Measure → Decide**

## Assessment objective

Design a practical AI workflow in which every stage produces an intermediate output that becomes structured input for the next stage. The workflow avoids a single oversized prompt and instead demonstrates controlled, traceable reasoning.

## Chosen campus observation

**“Students frequently leave classroom fans and lights ON after leaving the classroom.”**

# 01 WORKFLOW BLUEPRINT

The workflow is intentionally designed as a chain of dependent prompts. The AI does not jump directly from a problem to a solution; it first establishes what is known, separates assumptions, identifies likely causes, and only then designs a small test.

| | | | |
|---|---|---|---|
| **INPUT** | **1**<br>Problem<br>Decomposition | **2**<br>Root Cause<br>Analysis | **3**<br>Cause<br>Prioritization |
| **4**<br>Micro-Experiment<br>Design | **5**<br>Measurement<br>Framework | **6**<br>Final Action<br>Plan | **OUTPUT** |

## Design principles

| Principle | How it appears in the workflow |
|---|---|
| Traceability | Each step explicitly consumes a previous output. |
| No premature solutions | Diagnosis happens before intervention design. |
| Testability | The proposed intervention can succeed or fail. |
| Measurability | The final workflow defines metrics and a decision rule. |
| Responsible prompting | The AI is told not to invent experimental results. |

## 01 Problem Decomposition

Establish a precise problem definition before proposing solutions.

```
You are a campus innovation analyst.

Analyze the following real-world campus observation:

“Students frequently leave classroom fans and lights ON after leaving the classroom.”

Break this observation into:
1. The visible problem
2. The underlying problem
3. Who is affected
4. Who is responsible or involved
5. When and where the problem is most likely to occur
6. What resources may be unnecessarily consumed
7. What information is still missing

Do not suggest solutions yet. Focus only on understanding and defining the problem.

Present the result in a structured table.
```

### Expected intermediate output

A structured problem map containing the visible issue, underlying issue, affected stakeholders, likely context, resource impact, and missing information.

## 02 Root Cause Analysis

Turn the initial definition into testable explanations rather than assumptions.

```
Using the problem analysis from Step 1, perform a root-cause analysis.

Identify possible causes under these categories:
- Human behavior
- Process
- Environment
- Technology
- Communication
- Responsibility/ownership

For every possible cause, explain the mechanism by which it could lead to the problem.

Separate:
A. Direct causes
B. Contributing causes
C. Assumptions that need verification

Do not present solutions yet.
```

### Expected intermediate output

A categorized list of direct causes, contributing causes, and assumptions requiring verification.

## 03 Cause Prioritization

Select which suspected cause deserves testing first.

```
Using the root causes identified in Step 2, create a cause-priority matrix.

Score each cause from 1–5 for:
- Likelihood
- Potential impact
- Ease of testing

Calculate:
Priority Score = Likelihood × Impact × Ease of Testing

Rank all causes from highest to lowest priority.

Do not recommend a permanent solution. The purpose is to determine which causes should be tested
first.
```

### Expected intermediate output

A ranked cause-priority matrix using a transparent scoring formula.

## 04 Micro-Experiment Design

Design a small, low-cost experiment that can prove the intervention useful or ineffective.

```
Using the highest-priority cause from Step 3, design a small campus micro-experiment that can be
completed without expensive equipment.

The experiment must:
1. Have a clear hypothesis.
2. Use a baseline period.
3. Have an intervention period.
4. Change only one major variable.
5. Define what will be observed.
6. Be feasible for students to conduct.
7. Avoid collecting personal or identifying information.

Do not assume that the intervention will work. The experiment should be capable of proving the
intervention ineffective.
```

### Expected intermediate output

A baseline-versus-intervention micro-experiment with a falsifiable hypothesis.

## 05 Measurement Framework

Define how success will be evaluated without fabricating data.

```
Create a simple measurement framework for the micro-experiment.

Define:
1. Primary metric
2. Secondary metrics
3. Baseline measurement
4. Intervention measurement
5. Success threshold
6. Possible confounding factors
7. How the results should be interpreted

Use measurements that students can collect manually without specialized equipment.

Do not invent experimental results.
```

### Expected intermediate output

A measurement plan with primary/secondary metrics, a success threshold, confounders, and interpretation rules.

## 06 Final Action Plan

Convert the previous reasoning into a concise, decision-ready proposal.

```
Using the outputs from Steps 1–5, prepare a one-page campus micro-experiment proposal.

Include:
1. Project title
2. Problem statement
3. Hypothesis
4. Root cause being tested
5. Experimental design
6. Duration
7. Required resources
8. Measurement method
9. Success criterion
10. Risks and limitations
11. Decision rule after the experiment
12. Recommended next step

Clearly distinguish between:
- Known information
- Assumptions
- Measurements that still need to be collected

Do not invent experimental results.
```

### Expected intermediate output

A one-page proposal that can be reviewed by a campus team and used to decide whether the intervention should be tested at larger scale.

# 08 EXAMPLE FINAL RESULT

The following is the type of final deliverable produced by the workflow. It intentionally does not claim that the experiment has already been conducted.

## LAST-PERSON-OUT ENERGY EXPERIMENT

**Problem.** Classroom electrical equipment may remain ON after rooms become unoccupied.

**Hypothesis.** A simple exit reminder combined with explicit shutdown responsibility will reduce unnecessary operation.

**Root cause tested.** Forgetting to switch equipment OFF.

| Phase | Duration | Action |
|---|---|---|
| Baseline | 3 days | Observe normal classroom conditions without introducing the intervention. |
| Intervention | 3 days | Place an exit reminder and assign an existing class representative as the shutdown checker. |
| Comparison | After collection | Compare the incident rate during baseline and intervention periods. |

**Primary metric:** Number of observed empty classrooms with at least one unnecessary light/fan still operating.

**Example success criterion:** A reduction of at least 30% compared with baseline would indicate that the intervention is promising. This is a proposed threshold, not an observed result.

**Decision rule:** If the threshold is met, test the approach in additional classrooms. If it is not met, investigate other causes before scaling.

**Limitations:** A short experiment cannot establish long-term behavioral change; schedule differences and unusual campus events may affect the observations.

**Important evidence boundary**
**The AI workflow generates hypotheses, experiment structure, and decision criteria. Actual experimental results must come from real observations rather than AI-generated numbers.**

# 09 CONCLUSION

This workflow demonstrates a practical use of sequential prompting for a real-world campus problem. Instead of asking an AI system for a direct answer, the task is decomposed into six controlled stages. Each stage has a specific purpose, a defined prompt, and a clear intermediate output.

The workflow combines problem framing, qualitative reasoning, simple quantitative prioritization, experimental design, measurement planning, and decision-making. The final proposal is therefore traceable back to the original observation.

**Key takeaway**
**The value of the workflow is not merely the final answer. It is the sequence that makes the answer *structured, testable, measurable, and auditable*.**

## Assessment checklist

- ■ A clear real-world input is defined.
- ■ Six sequential prompt stages are specified.
- ■ Each stage has a distinct objective.
- ■ Intermediate outputs are passed to later stages.
- ■ The experiment has a baseline and intervention.
- ■ Metrics and a decision rule are defined.
- ■ The workflow explicitly prevents fabricated experimental results.